

NEUSE RIVER / HURRICANE FLORENCE 2018

Technical Documentation - Backend, Hydraulics, Flood Mapping and Impact Assessment

USGS 02089000 | Neuse River near Goldsboro, North Carolina | September 2018

1. Abbreviations and terminology

Short form	Meaning	Short form	Meaning
AOI	Area of Interest	NHDPlus HR	National Hydrography Dataset Plus High Resolution
AHPS	Advanced Hydrologic Prediction Service	NLCD	National Land Cover Database
CDS	Copernicus Climate Data Store	NLDI	Network-Linked Data Index
DEM	Digital Elevation Model	NOAA	National Oceanic and Atmospheric Administration
ERA5-Land	ECMWF land reanalysis	NSE	Nash-Sutcliffe Efficiency
ET0	Reference evapotranspiration	NWIS	National Water Information System
FIM	Flood Inundation Mapping	OSM	OpenStreetMap
GEE	Google Earth Engine	PBIAS	Percent Bias
GPKG	GeoPackage	Q	Discharge / streamflow
HAND	Height Above Nearest Drainage	QPE	Quantitative precipitation estimate
H-Q	Stage-discharge relationship	USGS	United States Geological Survey
KGE	Kling-Gupta Efficiency	UTC	Coordinated Universal Time
MRMS	Multi-Radar Multi-Sensor	WFO	Weather Forecast Office
NAVD88	North American Vertical Datum of 1988	WSE	Water Surface Elevation
NCEP	National Centers for Environmental Prediction	XGB	Extreme Gradient Boosting
NHC	National Hurricane Center	CRS	Coordinate Reference System

Short form	Meaning	Short form	Meaning
ECMWF	European Centre for Medium-Range Weather Forecasts	FAO-56	FAO reference method used for ET0
MAE	Mean Absolute Error	RMSE	Root Mean Square Error

1.1 Core project settings

Item	Project setting
USGS site	02089000 - Neuse River near Goldsboro, NC
Gauge coordinates	-77.9975, 35.3375
Gauge reference elevation	41.926 ft NAVD88 — NOAA/NWS GLDN7 FIM metadata for USGS 02089000
Published drainage area	2,399 mi ² ; watershed target approximately 6,232.005499 km ²
Daily mapping window	14-23 September 2018 (10 states)

“Branch” terminology: an H-Q branch means the rising or falling side of the stage-discharge curve. A workflow branch means a separate processing path, such as the NOAA FIM reference workflow. These are different meanings; the NOAA reference path is not an H-Q rising/falling branch.

2. Data collection, date ranges and data-gap handling

Each dataset has one clear role. The workflow keeps observation, rainfall, terrain and exposure sources separate and records the time period used for each major input.

Source / dataset	Project role	Date range / epoch
USGS NWIS	Florence discharge/stage, rating information and field measurements for site 02089000.	Event: 2018-09-10 to 2018-09-25; forcing: 2018-09-01 to 2018-09-26.

Source / dataset	Project role	Date range / epoch
USGS NLDI	Upstream watershed geometry; accepted only after gauge-location and drainage-area QC.	Current basin delineation service.
USGS NHDPlus HR (HU4 0302)	Flowlines, raw catchments and downstream routing used to create the upstream network and 37 modelling subcatchments.	no event time window required.
NOAA/NCEP MRMS	Hourly QPE rainfall for Florence and the 2015-2017 model-development archive.	Florence: 2018-09-01 to 2018-09-27; historical: 2015-05-10 to 2018-01-01; warmup: 2018-06-01 to 2018-09-27.
ERA5-Land via CDS	Antecedent soil/deep wetness and meteorology used for Physics V3.5 state and ET0.	Development: 2015-01-01 to 2018-01-01 (45-day antecedent windows); Florence state: 2018-06-01 to 2018-09-26.
2014 lidar DEM	Retained 1 m NAVD88 terrain used by the production flood mapper.	2014 terrain product.
Neuse River mainstem	Named Neuse River line used to build the longitudinal spatial WSE profile.	Hydrography reference for the Goldsboro reach.
NOAA FIM / AHPS	Small-domain GLDN7 reference branch; not used to generate full-AOI production flood maps.	no specific event acquisition date is required.
Copernicus EMSR311	Used only to recover the published Goldsboro analysis-AOI boundary.	Used only to recover the published Goldsboro analysis-AOI boundary.
WorldPop 2018 + NLCD 2016 + OSM	Population, land cover/imperviousness, buildings, roads and facilities for impact assessment; OSM is a current snapshot proxy as approximation of infrastructure exposure.	WorldPop 2018; NLCD 2016; OSM current at acquisition time.

2.1 Gap handling

Data stream	Data handling rule
Florence MRMS	Requires high hourly completeness and basin coverage; limited radar-only fallback is allowed within configured limits.

Data stream	Data handling rule
Historical MRMS	Checked for completeness, coverage, fallback hours and consecutive gaps; not freely interpolated.
Verified dry-hour gaps	Within the 2015-2017 historical model-development archive, a small number of missing MRMS hours inside calibration/validation periods were reviewed against surrounding observed basin rainfall. A zero value is used only when surrounding basin mean and basin maximum rainfall show that the missing hour was effectively dry. This is evidence-based gap repair, not generic interpolation.
H-Q / flood mapping	USGS field measurements are paired with gauge observations only within the defined time-matching tolerance. Discharge is converted to stage only within the observed range supported by the effective H-Q surrogate ; values outside that range are not extrapolated.

2.2 Effective H-Q Surrogate

We built the H-Q surrogate from USGS observed discharge and gauge-height data at station 02089000 during the Florence event. We paired discharge and stage by time, separated the observations into rising and falling limbs of the hydrograph, and built separate Q-to-H relationships. We then checked those relationships against available USGS field measurements before using them to convert modeled discharge into stage. At 5.1 Section we have a clear view of it.

The rainfall-gap policy is generic. A missing historical hour is repaired only when surrounding observed rainfall provides clear evidence that the hour was dry. Otherwise the gap remains a quality-control issue rather than being filled by interpolation.

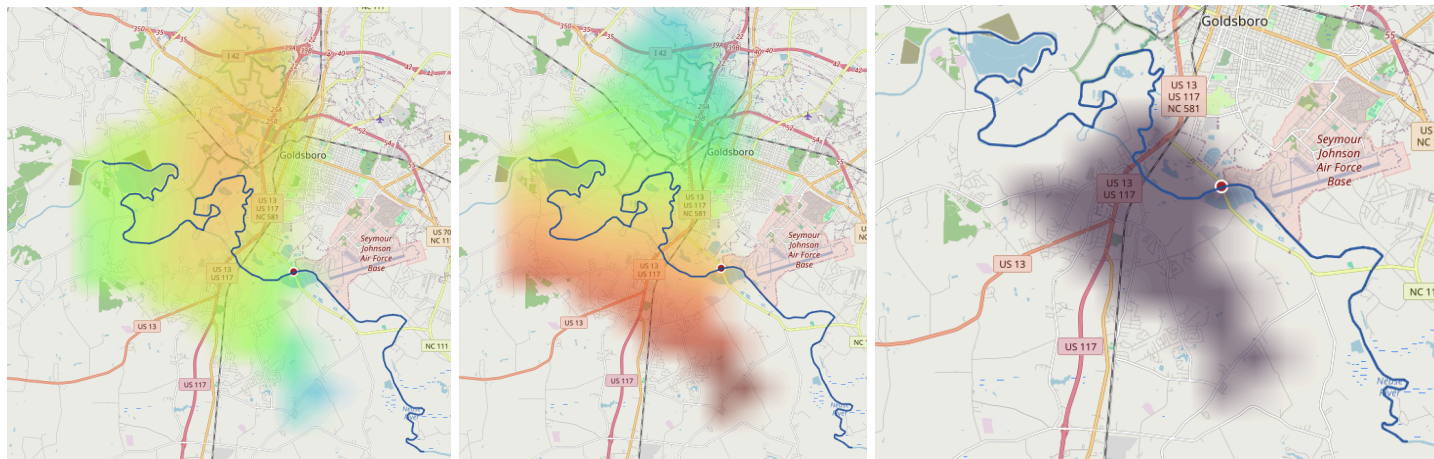


Figure: Rain fall proximity during experimental event window

3. Complete backend project flow - Part A: observations, domains and forcing

Step / module	Backend action -> output / purpose
0. Environment	environment_check.py verifies dependencies and external access before science processing starts.
1. USGS observations	Download event Q/H, current rating information, historical rating evidence and field measurements for H-Q checking. A field-measurement record therefore usually contains a timestamp, measured discharge QQ, and the gauge height HH at approximately the same time.
2. Effective H-Q surrogate	Build rising/falling event-effective Q-H curves, check field-measurement consistency and freeze a validated H-Q manifest.
3. Upstream watershed + AOI	NLDI builds the contributing watershed; prepare_reference_aoi.py separately recovers the local Goldsboro flood/impact AOI.
4. Florence MRMS rainfall	Acquire Sep 1-27 MRMS, standardize to EPSG:32618 at 1 km, apply watershed QC and create basin-hourly rainfall.
5. Florence forcing	Combine the finalized rainfall time series with USGS observations over warmup/evaluation windows.
6. Upstream network	Download NHDPlus HR HU4 0302, locate the gauge outlet, retain connected upstream flowlines/catchments and build routing edges.
6B. Subcatchments	Aggregate raw catchments into connected modelling units near 150 km ² ; the retained workflow expects 37.
6C. Rainfall zones + routing	Map rainfall cells to subcatchments and create common routing features shared by Physics V3.5 and ML V4.
Why this sequence is important The workflow first fixes the full upstream drainage and routing structure so rainfall is converted to discharge correctly before local flood extent and impacts are calculated.	

3. Complete backend project flow - Part B: model development, flood and impact

Step / module	Backend action -> output / purpose
7. Historical rainfall + Q	Build 2015-2017 basin/subcatchment rainfall and USGS discharge; apply only documented dry-hour repairs.

Step / module	Backend action -> output / purpose
8. Florence warmup rainfall	Build continuous June-Sep 2018 spatial rainfall so event prediction starts from a long warmup.
9. Physics event library	Create five 2015-2016 calibration blocks and three 2017 temporal-check blocks; Florence is excluded from this clean library command.
10. ERA5-Land state	Prepare soil wetness, ET0 and deep-state scaling for Physics V3.5.
11. Physics production configuration	Use the current source-controlled Physics V3.5 parameter file. The clean command does not run a new optimization, which keeps Florence prediction reproducible.
12. ML V4	Build exogenous rainfall/routing features, train 2015-2016, and select among ten fixed XGBoost candidates using 2017 validation blocks.
13-14. Florence prediction	Prepare Florence continuous state and produce Physics V3.5 and ML V4 discharge time series.
15-16. FIM + comparison	Download NOAA FIM reference data and compare USGS/Physics/ML Q and branch-derived stages.
17-18. Flood mapping	Use the production 1 m lidar DEM and named Neuse mainstem; Q -> H-Q stage -> gauge WSE -> longitudinal WSE -> connected flood extent/depth.
19-20. Impact assessment	Acquire WorldPop, NLCD and OSM exposure; summarize population, buildings, roads, facilities, land cover and imperviousness.
21-22. Dashboard	Build read-only interactive visualization and date-specific impacted-road vectors; no model retraining occurs.
Validation language ML V4 uses 2015-2016 for training and 2017 event windows for internal model selection/validation. Florence is excluded from ML fitting and candidate selection. For Physics V3.5, the current production parameter set was finalized during earlier project development in which Florence observed discharge was part of the historical selection information. Therefore the Physics Florence output is described as a model application/comparison result, not as an independent validation result.	

3.1 Calibration and temporal-check event library

The clean development library used five calibration blocks from 2015–2016 and three temporal-check blocks from 2017. Each block included pre-event conditions, **a 72-hour warm-up, the rising limb and peak, and the recession period**. Physics V3.5 used **MRMS rainfall, ERA5-Land wetness/state, ET0, 37-subcatchment routing, and USGS observed discharge** for

calibration. The five earlier blocks were used to optimize parameters, while the three 2017 blocks were used only for temporal checking. Florence 2018 was excluded and applied afterward as the final event case.

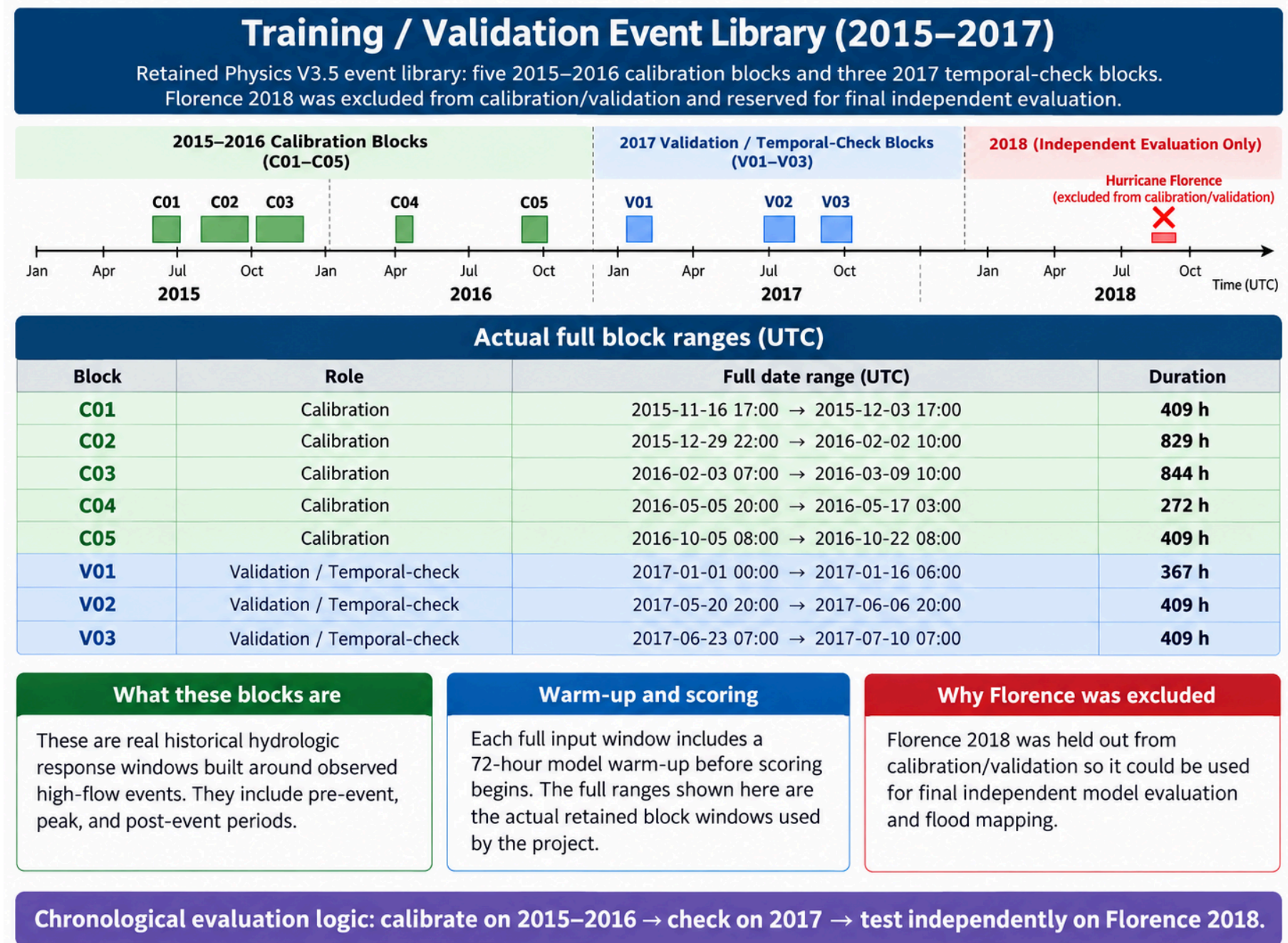


Figure 3.1: Event Library Calibration Blocks

4. Catchment, upstream watershed, subcatchments and Goldsboro AOI

4.1 Upstream watershed calculation - plain-language explanation

The backend asks USGS NLDI for the land area that drains to USGS 02089000. It checks that the gauge location is correct and that the calculated drainage area is close to the USGS published value of 2,399 mi². If one supported NLDI request type is unavailable, the script tries another NLDI option and accepts it only after the same quality checks. The local Goldsboro map area is never substituted for the full upstream watershed.

4.2 Modelling catchments / subcatchments

NHDPlus HR divides the upstream watershed into many raw catchments connected by the river network. The project groups these into 37 connected modelling subcatchments. Each subcatchment keeps a downstream path toward USGS 02089000, so rainfall falling far upstream can be delayed and routed through the network before it contributes to discharge at Goldsboro.

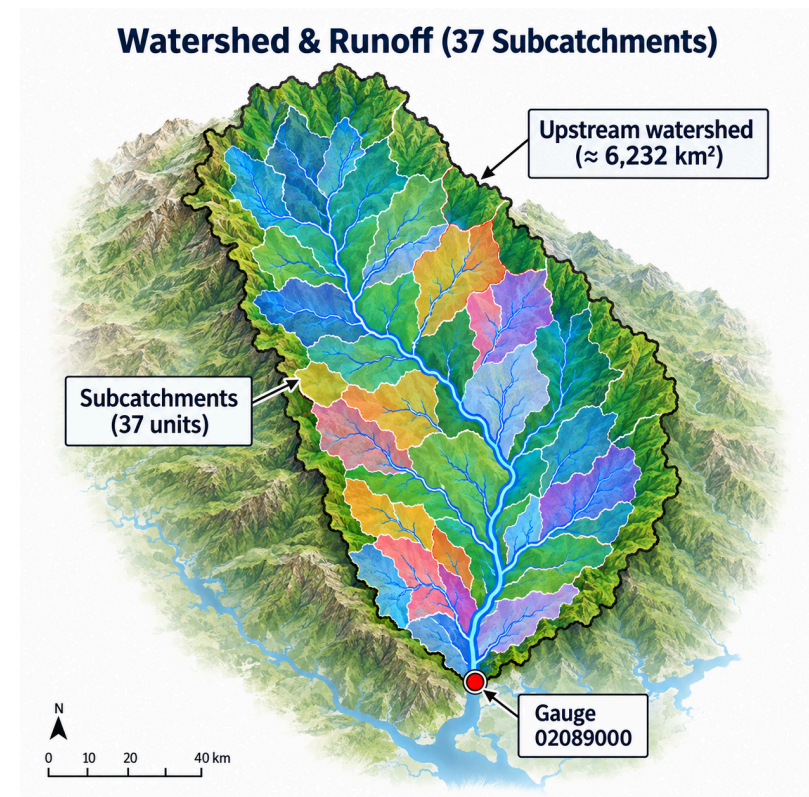
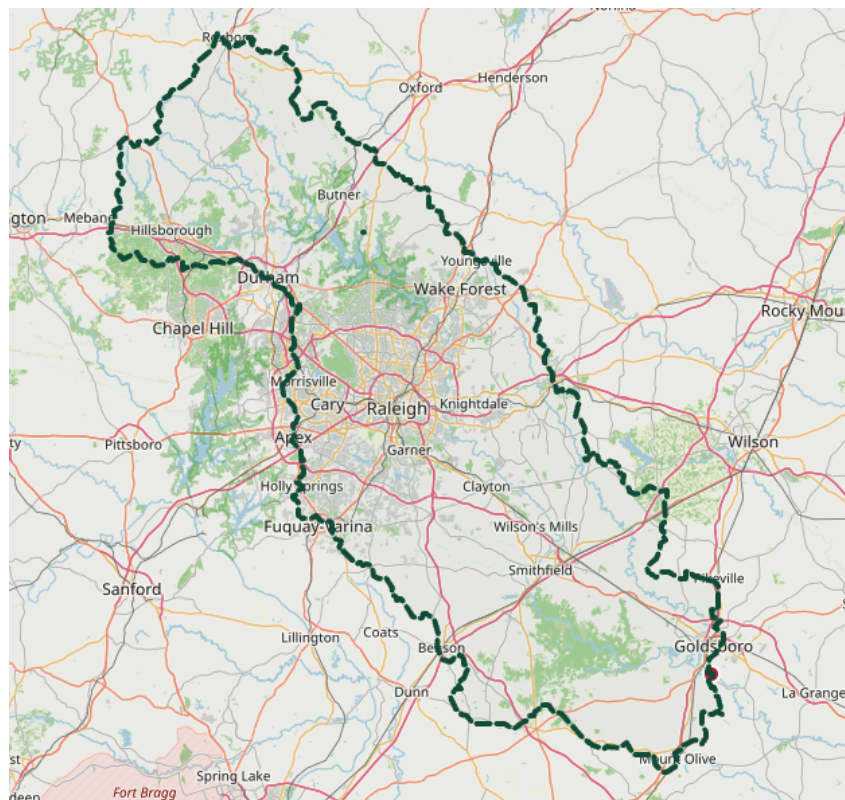


Figure 4.1(Left): Upstream watershed / catchment contributing to USGS 02089000.

Figure 4.2(Right): Conceptual watershed and 37 connected subcatchments draining toward gauge 02089000.

4.3 Goldsboro flood-mapping AOI

The local Goldsboro AOI is used for terrain, flood and impact products; it is not the upstream rainfall basin. This separation prevents a map viewport from being mistaken for the hydrologic contributing area.

4.4 Downstream mainstem, gauge WSE and production DEM

The hydrologic watershed is the contributing area upstream of USGS 02089000. Flood mapping, however, uses the Neuse River mainstem both upstream and downstream of the gauge within the local Goldsboro AOI. This allows WSE to change along the river rather than applying one gauge elevation as a flat plane.

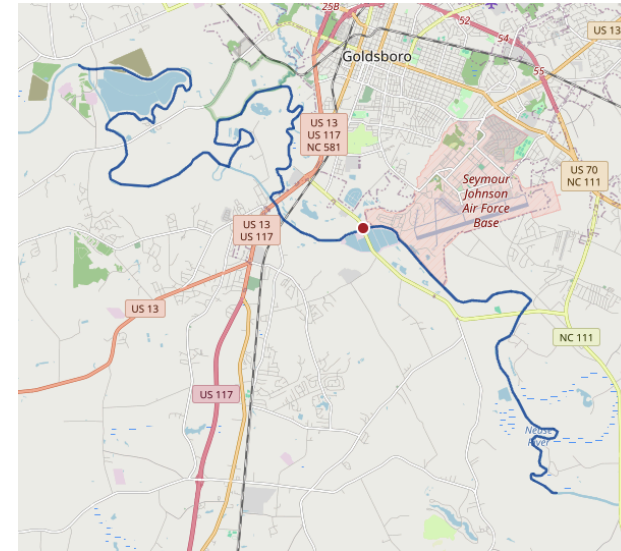
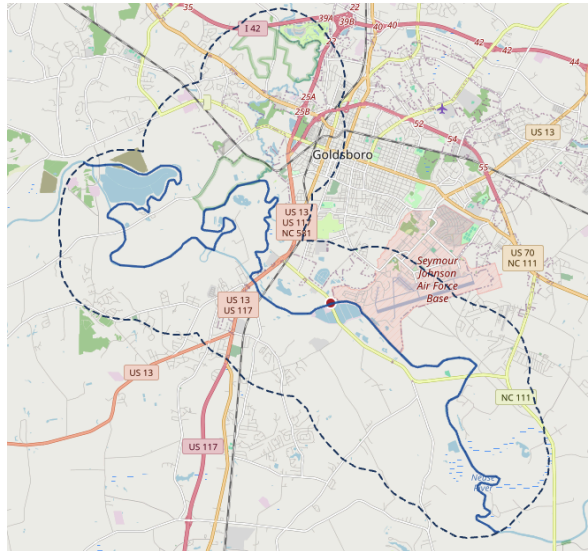


Figure 4.3(Left): Goldsboro flood-mapping / modelling AOI around the gauge.

Figure 4.4(Right): Neuse River mainstem through Goldsboro with the gauge shown at USGS 02089000.

4.5 Gauge WSE

Quantity	Project relation
Stage H	Local gauge height above the station datum.
Gauge WSE	$\text{WSE Gauge (ft NAVD88)} = 41.926 \text{ ft} + H \text{ (ft)}.$
Mapper units	Gauge WSE is converted internally to metres NAVD88 before comparison with the DEM.

4.6 Production DEM

The production flood mapper uses **output/terrain/goldsboro_dem_2014/goldsboro_dem_2014_1m_navd88_m.tif**, an original unconditioned 1 m lidar DEM in metres NAVD88.

Why it matters: The high-resolution DEM decides which local floodplain cells lie below the spatial water surface.

5. Effective H-Q surrogate, H-Q branch and daily hydraulic evolution

5.1 Effective Florence H-Q surrogate

An authoritative historical rating artifact exactly matching the Florence event was not available. The project therefore builds an event-effective H-Q relationship from paired USGS discharge and gauge-height observations, represents the rising and falling sides separately, and checks the resulting curves against USGS field measurements before they are used by the flood mapper.

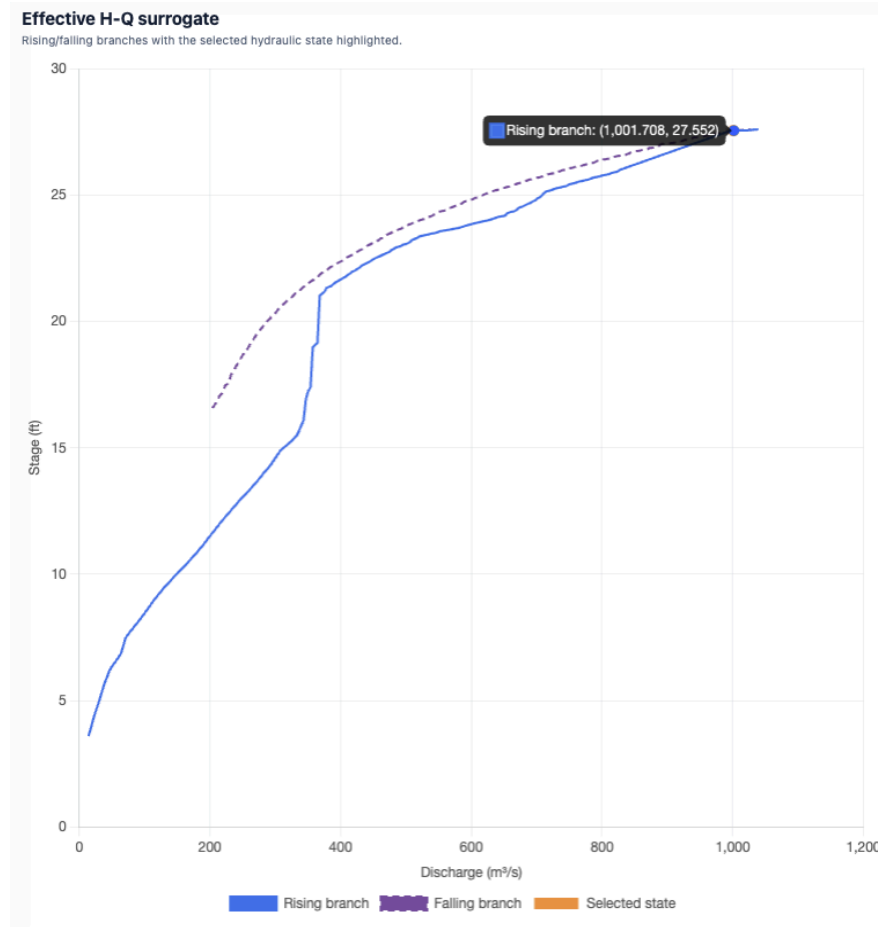
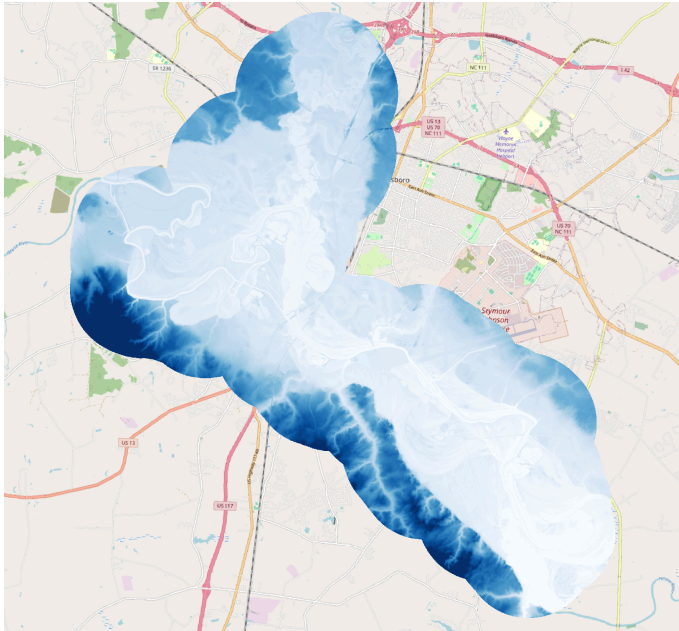


Figure 4.6(Left): AOI DEM For Production use(Dark Blue - Low Reach Area, Light Blue - High reach Area)

Figure 5.1(Right): Effective H-Q relationship.

5.2 H-Q branch - Workflow branch (Simple explanation)

An H-Q branch means which side of the flood the river is on. Rising branch = the river is still increasing toward its discharge peak. Falling branch = the river has passed the peak and is receding. The same discharge can correspond to a slightly different stage on the way up versus the way down because channel and floodplain storage are different. The mapper checks when the selected daily discharge occurs and uses the matching H-Q curve. It does not average the two curves, and it does not extrapolate outside the supported discharge range.

Simple example: imagine the river reaches the same discharge once while rising and again while receding. If the two curves in the H-Q graph give different stages for that same Q, the mapper uses the stage from the curve that matches the event state. This preserves the observed flood hysteresis rather than forcing one single curve.

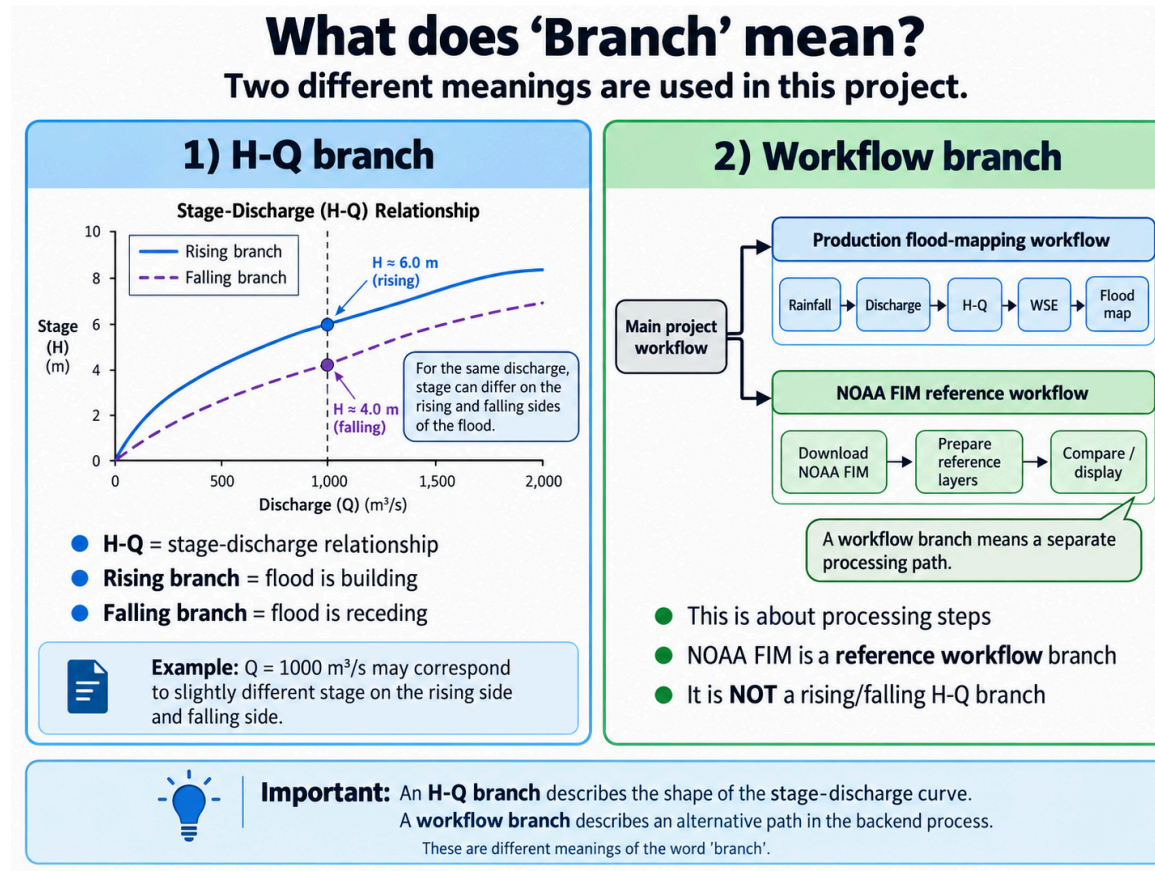


Figure 5.2: Two meanings of “branch”: an H-Q branch is the rising/falling side of the stage-discharge relationship; a workflow branch is a separate processing path such as NOAA FIM.

5.3 Daily hydraulic evolution

Daily discharge controls branch-derived stage; stage plus the gauge datum gives gauge WSE. As discharge rises, WSE rises and more low terrain can connect to the river. During recession the reverse occurs.

Daily hydraulic evolution

Discharge, stage, gauge WSE and HAND-equivalent threshold · 14–23 Sep.

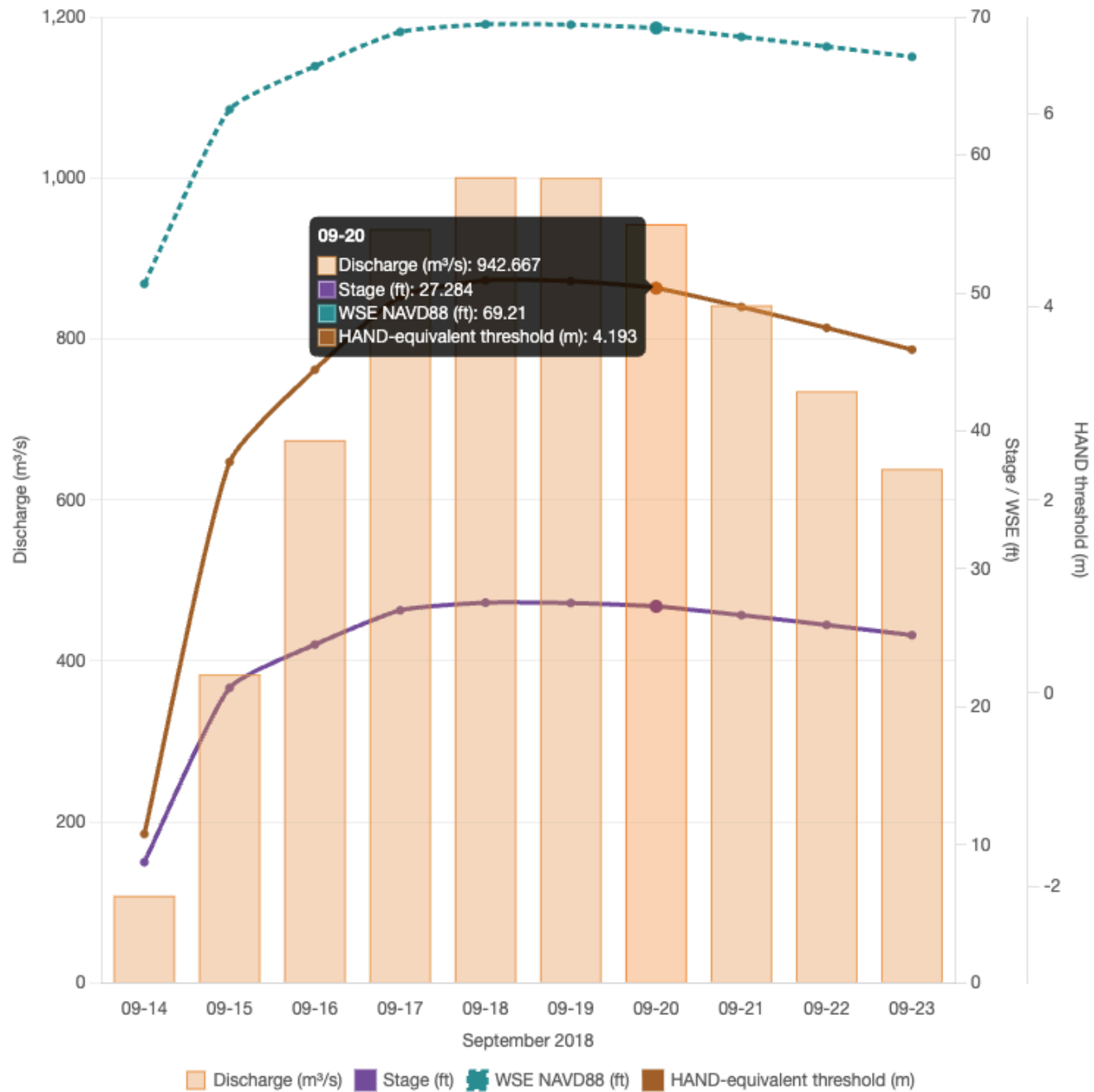


Figure 5.3: Daily discharge, stage, gauge WSE and HAND-equivalent threshold, 14-23 September 2018.

5.4 Rainfall -> discharge -> hydraulic response

MRMS rainfall is the forcing, but discharge is the delayed basin response after infiltration/storage, runoff production and channel routing. Therefore rainfall can decrease before the river peaks, and flood extent can remain large after daily rainfall approaches zero.

The daily plot is a readable summary of 24-hour MRMS accumulation; the models themselves use the continuous hourly/spatial archive.

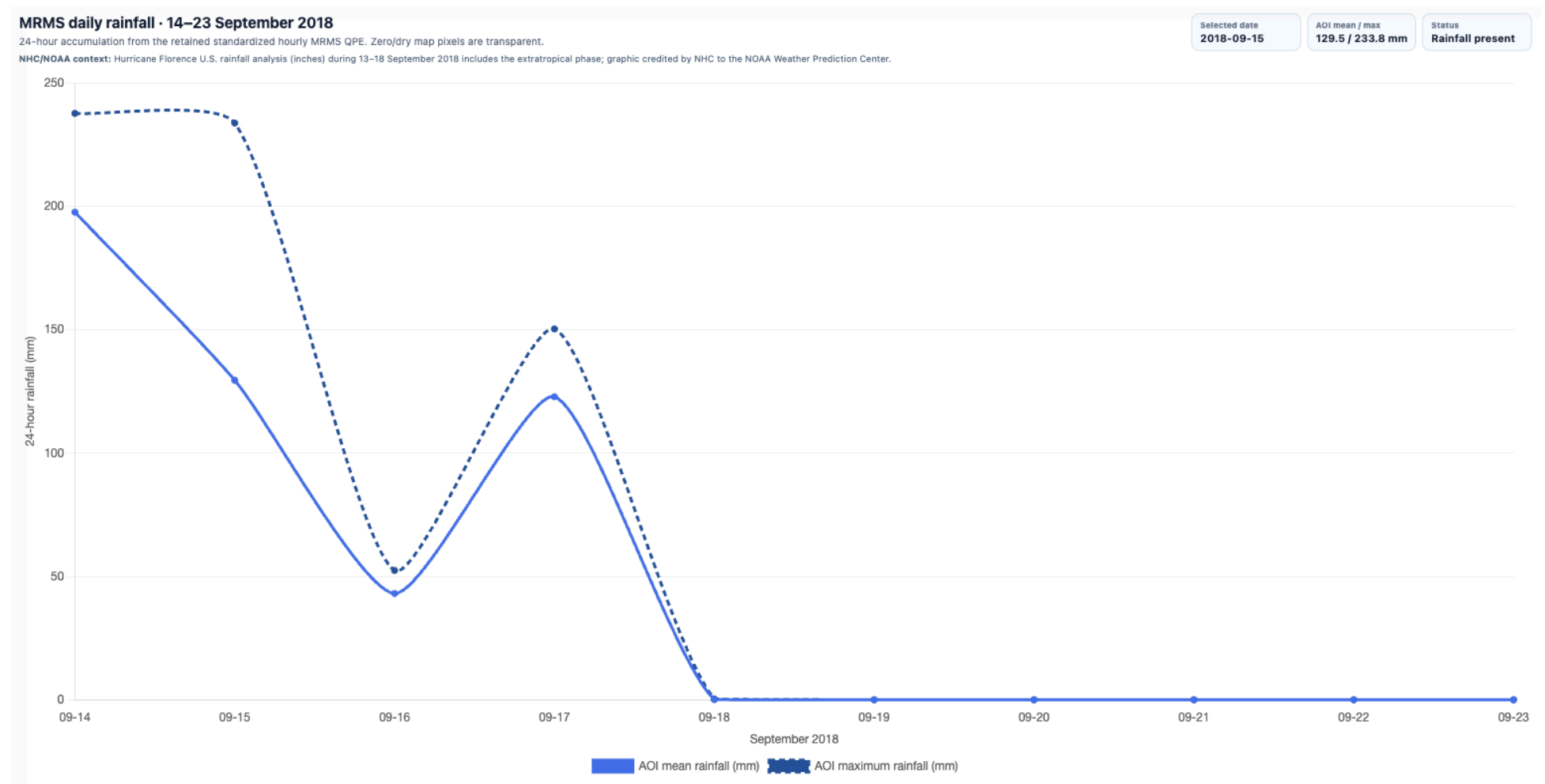


Figure 5.4: Daily MRMS rainfall summary, 14-23 September 2018.

Sequence summary: rainfall -> catchment storage/runoff -> subcatchment/channel routing -> discharge Q at Goldsboro -> branch-aware H - Q stage -> gauge WSE -> spatial WSE -> terrain comparison -> flood extent/depth.

6. Discharge models - Physics V3.5 and ML V4

Both models predict discharge at USGS 02089000; afterward they use the same H-Q conversion and retained terrain/WSE mapper. The table emphasizes the differences that matter scientifically and reproducibly.

Item	Physics V3.5	ML V4
Type	Semi-distributed process-based rainfall-runoff + routing.	XGBoost regression with exogenous rainfall/routing features.
Development	5 calibration blocks (2015-2016) + 3 temporal-check blocks (2017).	Train through 2016; 2017 used as model-selection/validation holdout.
Training / selection	The clean workflow does not re-optimize; it uses the current source-controlled Physics V3.5 production parameter file.	<code>train_ml_v4.py</code> trains and compares 10 predefined XGBoost hyperparameter configurations using 2015–2016 training data and 2017 temporal-validation events, then retains the best-performing acceptable configuration for Florence prediction.
Model / configuration file	<code>src/models/physics/retained/physics_v3_5_round2_candidate11_parameters.json</code>	<code>output/models/ml_v4/ml_v4_model.joblib</code>
Prediction code	<code>predict_physics_v3_5_florence_continuous.py</code>	<code>predict_ml_v4_florence.py</code>
Inputs	Spatial rainfall, forcing, ERA5-Land soil/ET0/deep state, routing.	Spatial rainfall + routing features + frozen feature metadata; no observed-Q lag/recursive-Q.
Antecedent state	Explicit June-Sep continuous state and ERA5-Land wetness/ET0.	Rainfall-history and routing/rolling features over continuous Florence feature windows.
Output	<code>output/models/physics_final_florence/physics_v3_5_florence_2018_predictions.csv</code>	<code>output/models/ml_v4_florence/ml_v4_florence_2018_predictions.csv</code>

Observed driver: USGS discharge/stage are formatted into the same mapping input structure and passed through the same flood mapper. In this sentence “observed driver” means a data source, not an H-Q branch. The H-Q branch still means rising or falling.

6.1 Scientific logic, equations and exact model-file locations

Physics V3.5 - moisture-dependent preferential interflow

Physics V3.5 uses a **semi-distributed conceptual rainfall-runoff method** across 37 subcatchments. It represents soil storage, evapotranspiration, quick runoff, infiltration, slow drainage/baseflow, preferential interflow and downstream channel routing. The V3.5 change makes preferential interflow strongest when soil is dry and progressively weaker as soil becomes wetter. This was chosen because one constant bypass fraction was too rigid across different wetness conditions.

Scientific method and alternatives: the production Physics V3.5 path is a semi-distributed conceptual rainfall-runoff model with subcatchment storage, evapotranspiration, fast/slow runoff components, moisture-dependent preferential interflow and downstream channel routing. It was retained because it preserves physical interpretation while representing spatial rainfall and travel time. Methods not used as the production Physics discharge path are: a fully lumped model (too little spatial detail), a full 2-D hydrodynamic solver (a different, much heavier hydraulic approach rather than the project's discharge-prediction method), and a pure ML replacement (kept separately as ML V4 so physics and data-driven predictions can be compared).

Relation	Meaning
$\text{dryness} = \text{clip}(1 - S/C, 0, 1)$	S = soil storage; C = soil capacity.
$\text{f_pref_effective} = \text{f_pref_max} \times \text{dryness}$	Effective preferential fraction decreases as relative soil wetness increases.
$\text{interflow_input} = \text{infiltration} \times \text{f_pref_effective}$	Only the moisture-dependent fraction of infiltration enters preferential interflow.

ML V4 - causal exogenous features

ML V4 uses XGBoost regression. Its features are built only from exogenous rainfall, routing and time information: basin rainfall mean/variability/max/P90, area above rainfall-intensity thresholds, rolling rainfall totals, near/middle/far routing zones, routed-rainfall indices and seasonal/time terms. The feature metadata explicitly records no observed-Q lags, no future rainfall and no recursive predicted-Q state.

Key file locations:

Physics production parameter file: src/models/physics/retained/physics_v3_5_round2_candidate11_parameters.json.

Physics Florence prediction: src/models/physics/predict_physics_v3_5_florence_continuous.py.

ML feature preparation: src/models/ml/prepare_ml_v4_features.py.

ML training/validation: src/models/ml/train_ml_v4.py.

Frozen ML model: output/models/ml_v4/ml_v4_model.joblib.

Physics method summary: 37-subcatchment semi-distributed rainfall-runoff modelling uses ERA5-Land antecedent wetness, hourly ET0, soil storage, saturation-dependent quick runoff, infiltration, drainage/baseflow, moisture-dependent preferential

interflow and channel routing. This method is used because it keeps the rainfall-to-discharge process physically interpretable while still representing spatial rainfall and travel time across the watershed.

ML V4 development summary: `prepare_ml_v4_features.py` builds the 2015-2017 dataset from 37-subcatchment MRMS rainfall, routing features and **USGS Q as the target**. Training uses 2015-2016; 2017 event windows are the validation/model-selection holdout. `train_ml_v4.py` evaluates ten fixed XGBoost configurations with NSE, KGE, PBIAS, peak error and peak timing. Saved results include `ml_v4_model.joblib`, `ml_v4_candidate_metrics.csv`, `ml_v4_validation_predictions.csv`, `ml_v4_validation_event_metrics.csv`, `ml_v4_feature_importance.csv` and `ml_v4_metadata.json`.

6.2 ML V4 training strategy - exact project implementation

The ML workflow is time-aware and deliberately avoids information leakage. The following table describes the actual implementation in `prepare_ml_v4_features.py`, `train_ml_v4.py` and `predict_ml_v4_florence.py`.

Training aspect	Actual ML V4 implementation
Data preparation	37-subcatchment MRMS rainfall is converted into basin mean, spatial standard deviation, spatial maximum/P90, area-above-intensity fractions, short/long rolling rainfall totals, near/middle/far routing-zone rainfall and routed-rainfall indices. Observed USGS discharge is used only as the supervised target during development.
Boosting method	XGBRegressor gradient-boosted decision trees.
Optimization objective	objective = reg:squarederror, evaluation metric = RMSE, tree_method = hist. High-flow sample weights emphasize the upper discharge quantiles so flood peaks are not dominated by ordinary low-flow hours.
Hyperparameter tuning	learning_rate (0.020–0.035), n_estimators (700–900), min_child_weight (8–20), subsample (0.85–0.90), colsample_bytree (0.80–0.85), L2/L1 regularization, target transform (raw/sqrt/log1p) and high-flow weighting scheme.
Candidate selection	Candidates are ranked with a robust score using validation NSE, KGE, PBIAS, peak error and peak-timing error, including pooled validation performance plus event-level robustness. Quality gates are also enforced.
Recursive forecasting	Not used. predicted Q is never fed back as a state variable for the next prediction.
Direct forecasting	Used. Each hourly discharge prediction is produced directly from the prepared exogenous feature vector for that hour. No observed-Q lag, no future rainfall and no recursive predicted-Q input are used.

7. Flood mapping: rainfall -> discharge -> stage -> WSE -> terrain -> flood

The conceptual figure below is for mixed technical/non-technical audiences.

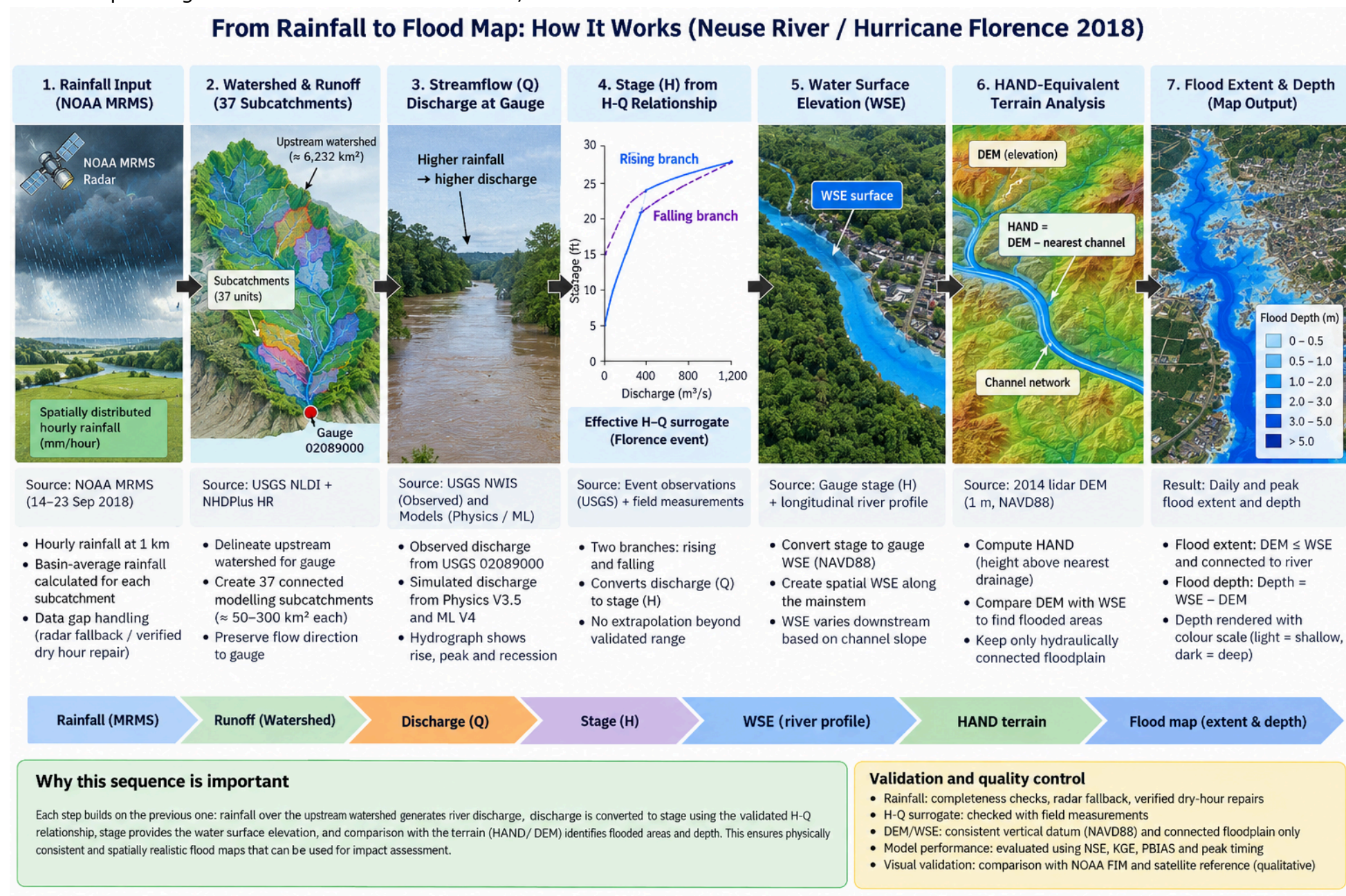


Figure 7: Conceptual end-to-end project workflow. It shows rainfall, subcatchments, discharge, H-Q stage, WSE, terrain/HAND interpretation and flood depth. The exact production calculation is listed below.

7.1 Exact production flood logic

- Select the daily maximum discharge from the chosen source: USGS observed, Physics V3.5 or ML V4.
- Check whether that daily state occurs on the rising or falling side of that source hydrograph.
- Convert discharge Q to stage H using the matching H-Q curve.
- Add the 41.926 ft NAVD88 gauge datum to obtain gauge WSE.
- Create a longitudinal WSE profile along the named Neuse River mainstem.
- Compare every valid 1 m DEM cell with the spatial WSE. A cell can be wet only when ground elevation is at or below the water surface.
- Keep only wet cells connected to the mainstem seed corridor; remove isolated low areas that are not hydraulically connected.
- Write flood extent and flood depth. For a flooded cell: depth = spatial WSE - DEM.

7.2 HAND, DEM and WSE - what is actually calculated

Conventional HAND is $\text{HAND}(x) = \text{DEM}(x) - \text{elevation of the nearest drainage reference}$. In this project, HAND is useful for understanding relative floodplain terrain, but the production flood calculation itself compares absolute spatial WSE with the lidar DEM and then applies mainstem connectivity. The dashboard HAND-equivalent value is therefore an interpretation aid, not a single threshold applied uniformly across the whole AOI.

Low, river-connected terrain is reached before higher ground as WSE rises. The following section explains how the numerical depth raster is calculated and then converted to blue colours for the dashboard.

7.3 Flood-depth calculation and colour rendering

Quantity	Meaning in the production / dashboard workflow
DEM(x)	Ground elevation at cell x in metres NAVD88.
WSE(x)	Spatial water-surface elevation at cell x in metres NAVD88.
Flood test	Cell is a candidate when $\text{DEM}(x) \leq \text{WSE}(x)$; only candidates connected to the mainstem seed remain flooded.
Depth value	$\text{Depth}(x) = \max(0, \text{WSE}(x) - \text{DEM}(x))$ inside the connected flood mask.
Raster storage	Depth is written as float32 metres; dry valid cells are 0 and no data is stored separately. In QGIS tools it is easier to understand.

Flood quantity	Project calculation
Gauge WSE	41.926 ft NAVD88 + branch-derived stage.

Flood quantity	Project calculation
Spatial WSE	Gauge WSE adjusted by the robust fitted longitudinal mainstem profile.
Candidate extent	DEM <= spatial WSE.
Final extent	Candidate wet cells connected to the mainstem seed corridor.
Depth	$\max(0, \text{spatial WSE} - \text{DEM})$ inside the connected extent.

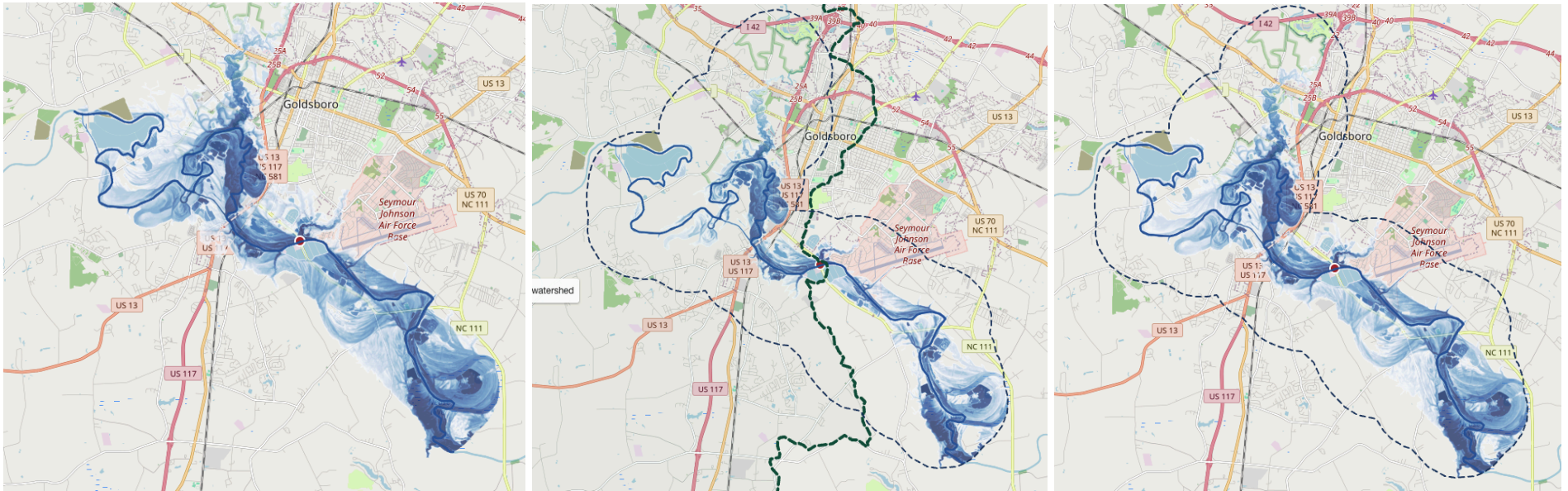


Figure 7.2: Actual project flood-depth map reference over Goldsboro. Darker blue areas represent greater calculated water depth; transparent areas are dry or outside the valid flood-depth display.

8. Impact assessment

WorldPop 2018 is used for population, NLCD 2016 for land cover and imperviousness, and OSM for buildings, roads and critical facilities. Population and NLCD use fractional raster overlap; roads are clipped to the flood geometry.

8.1 NLCD 2016 land-cover classes - text reference

Code	Class	Code	Class
11	Open Water	43	Mixed Forest
21	Developed, Open Space	52	Shrub/Scrub

Code	Class	Code	Class
22	Developed, Low Intensity	71	Grassland/Herbaceous
23	Developed, Medium Intensity	81	Pasture/Hay
24	Developed, High Intensity	82	Cultivated Crops
31	Barren Land	90	Woody Wetlands
41	Deciduous Forest	95	Emergent Herbaceous Wetlands
42	Evergreen Forest		

Scientific interpretation: water depth is the vertical gap between the spatial water surface and the local terrain elevation.

Exposure note: WorldPop 2018 and NLCD 2016 are event-era proxy datasets. OSM is currently at acquisition time unless replaced with a historical extract. The impact outputs should therefore be interpreted as flood-exposure estimates based on those source vintages.

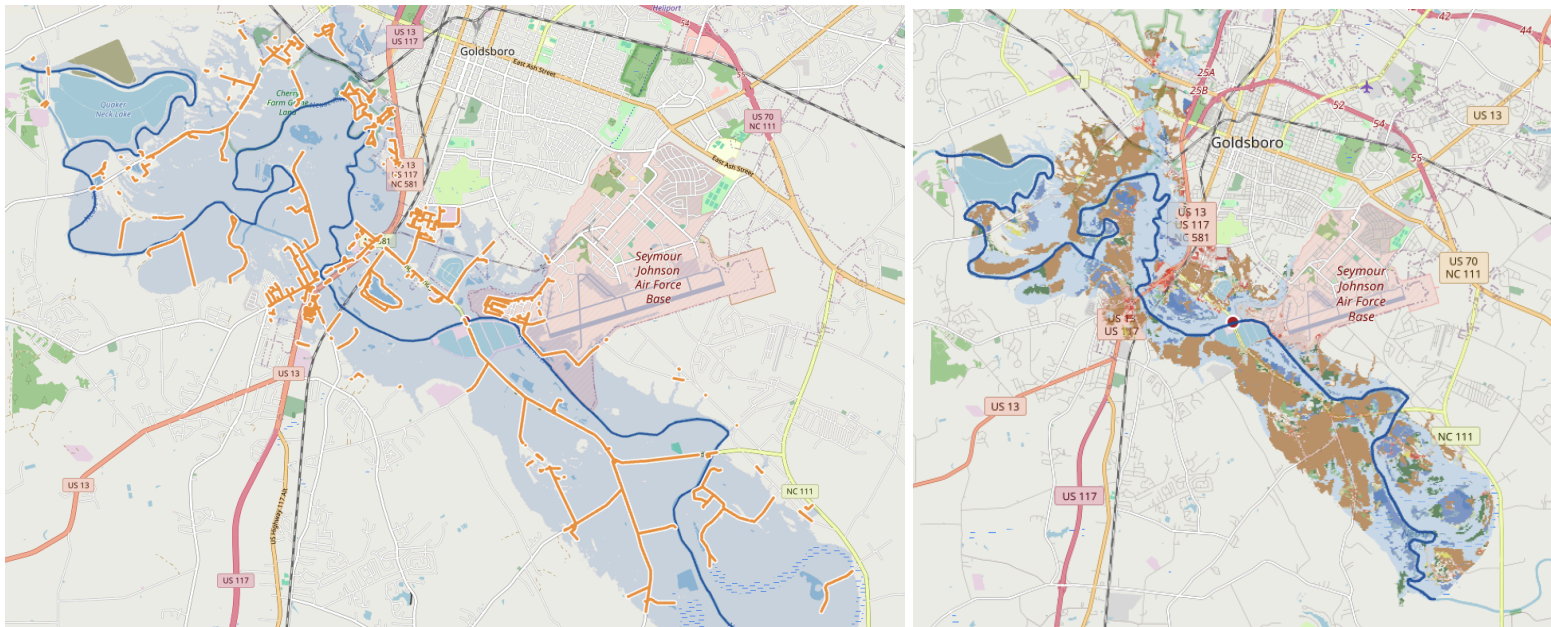


Figure 8(Left): Impacted Roads (Orange colored line)

Figure 8(Right): Impacted Land Cover

8.2 Observed vs Physics V3.5 vs ML V4 - Florence comparison results

The comparison below uses the accepted daily Florence reference table (14–23 September 2018).

Date	USGS observed Q (m ³ /s)	Physics V3.5 Q (m ³ /s)	ML V4 Q (m ³ /s)
2018-09-14	268.44	108.01	89.05
2018-09-15	376.61	383.05	169.54
2018-09-16	478.55	674.17	455.79
2018-09-17	826.85	936.69	823.65
2018-09-18	1039.23	1001.03	897.80
2018-09-19	1036.40	1000.12	891.27
2018-09-20	942.95	942.67	760.78
2018-09-21	792.87	841.70	743.65
2018-09-22	637.13	734.97	607.94
2018-09-23	509.70	638.50	586.17
Comparison summary	USGS observed	Physics V3.5	ML V4
Daily Q MAE, Sep 14–23 (m ³ /s)	—	82.26 (better than ML)	103.60
Daily stage MAE, Sep 14–23 (ft)	—	1.030(better than ML)	2.011
Observed-peak date	2018-09-18	2018-09-18	2018-09-18
Q on observed-peak date (m ³ /s)	1039.23	1001.03	897.80
Stage on observed-peak date	27.600 ft	27.552 ft	27.040 ft

Final Interpretation: Our Physics V3.5 version is closer to observed discharge and stage on average than ML V4 - while both models reproduce the rise, peak and recession pattern.